Supplementary Information

Observation of kagome-like bands in two-dimensional semiconducting Cr_8Se_{12}

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Supplementary Notes

Part 1. Characterization on the crystal structure and electronic properties of 2D Cr_2Se_3 with the 5-atomic-layer thickness

Detailed scanning probe microscopy characterizations were performed to determine the crystal structure and electronic properties of 2D Cr_2Se_3. The atomic-resolved STM image of 2D Cr_2Se_3 is displayed in Supplementary Figure 1a with the primitive unit cell marked by an orange rhombus. The lateral lattice constant is measured to be $|\boldsymbol{a}_0| = 0.370 \pm 0.003$ nm.

Additionally, as shown in Supplementary Figure 1b, we also conducted frequency shift spectra characterization on both 2D Cr_2Se_3 and the graphite surface. By measuring the tip height difference between the turning points of the two frequency shift spectra, the physical thickness of 2D Cr_2Se_3 is estimated to be around 0.945 nm, which is comparable to the STM apparent thickness of 1.03 nm revealed in Fig. 2a. Considering that the growth condition, flake thickness, and in-plane lattice constant are similar to the 5-atomic-layer Cr_2Se_3 previously reported in reference 1, we determined the crystal structure of thinner flake in Fig. 2a as the one displayed in Supplementary Figure 1c.

We also demonstrated the epitaxial mode of 2D Cr_2Se_3 on the graphite substrate using atomic-resolved nc-AFM/STM characterizations. The nc-AFM image of 2D Cr_2Se_3 and the STM image of the nearby graphite substrate are displayed in Supplementary Figure 1d and Supplementary Figure 1e, with their Fourier transformed images displayed in

50 S1f and S1g, respectively. It is notable that the lattice of 2D Cr_2Se_3 aligns with that of
51 the graphite substrate, demonstrating that the 2×2 unit cell of 2D Cr_2Se_3 is
52 commensurate with the 3×3 unit cell of the graphite. This is also clearly demonstrated
53 in the LEED pattern shown in Fig. 3c and reveals the epitaxial growth mode of 2D
54 Cr_2Se_3 on the graphite. The epitaxial growth mode promises the single crystalline nature
55 of the chromium selenide flakes and enables the ARPES characterization on its
56 electronic structures.
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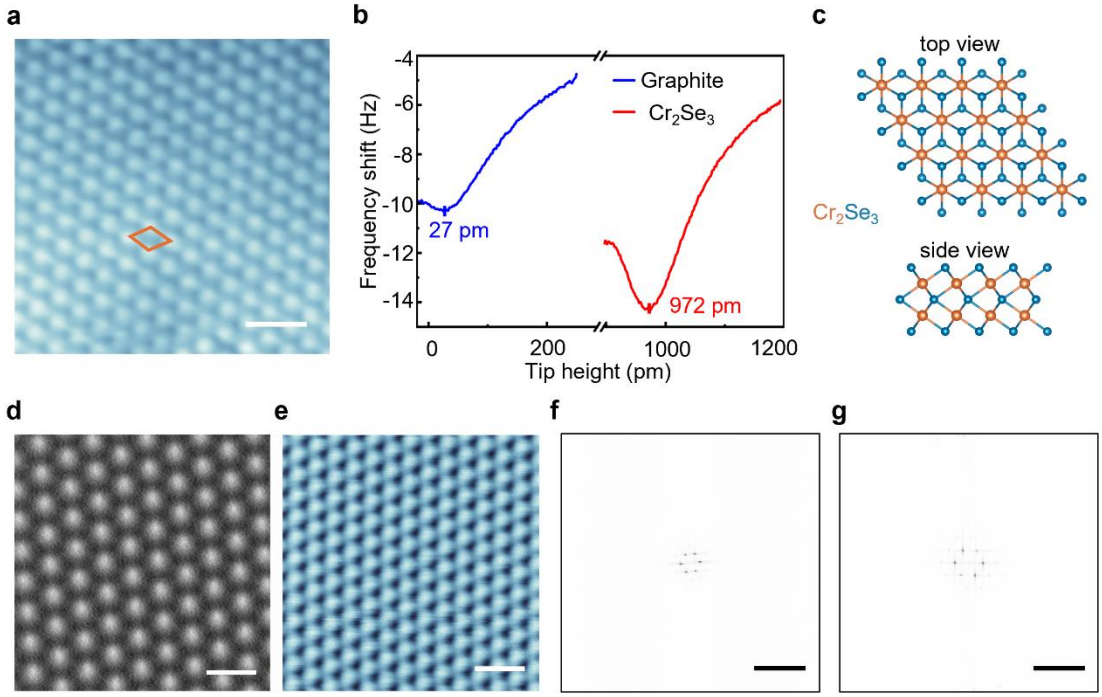
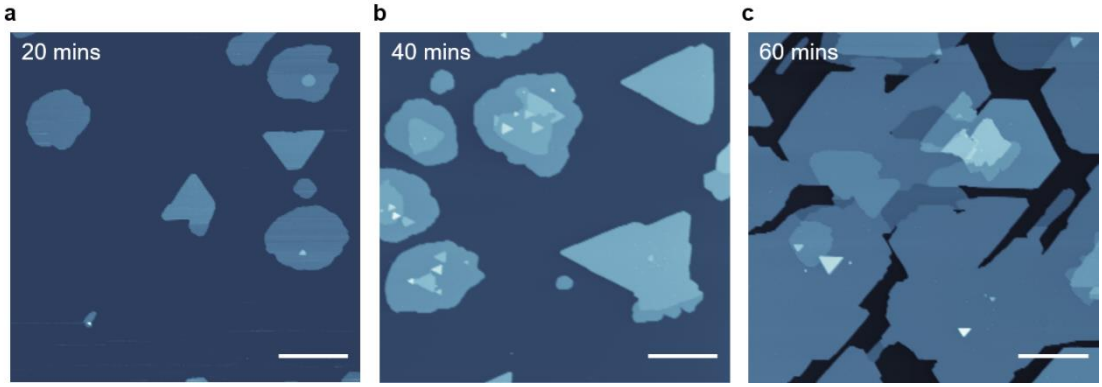


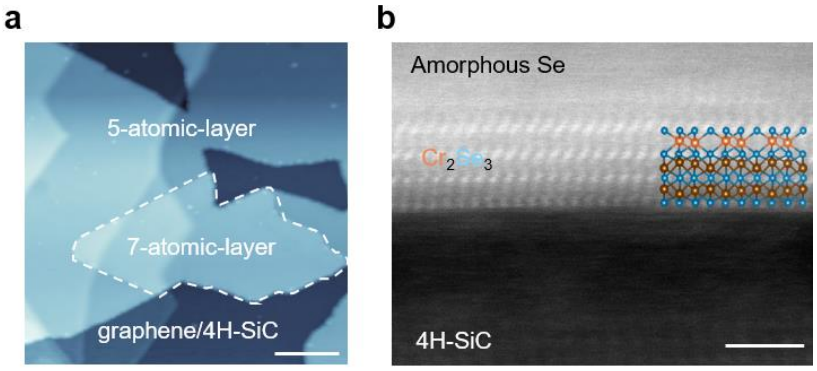
Supplementary Figure 1. The crystal structure of 2D Cr_2Se_3. (a) Atomic-resolved STM image of 2D Cr_2Se_3 with the unit cell marked by an orange rhombus. (V_s = 0.06 V, I = 1nA. T = 77 K.) Scale bar, 1 nm. (b) Frequency shift spectra acquired on both 2D Cr_2Se_3 and the graphite surface with the turning points marked by vertical lines. (c) Schematic crystal structure of 2D Cr_2Se_3. (d) The nc-AFM image a 2D Cr_2Se_3 flake. Scale bar: 1 nm. (e) The STM image of the graphite substrate beside the specific 2D Cr_2Se_3 flake in Supplementary Figure 1d. Scale bar: 0.6 nm. (f, g) Fourier transformed images of Supplementary Figure 1d and Supplementary Figure 1e, respectively. Scale bars: 17 nm^{-1}.



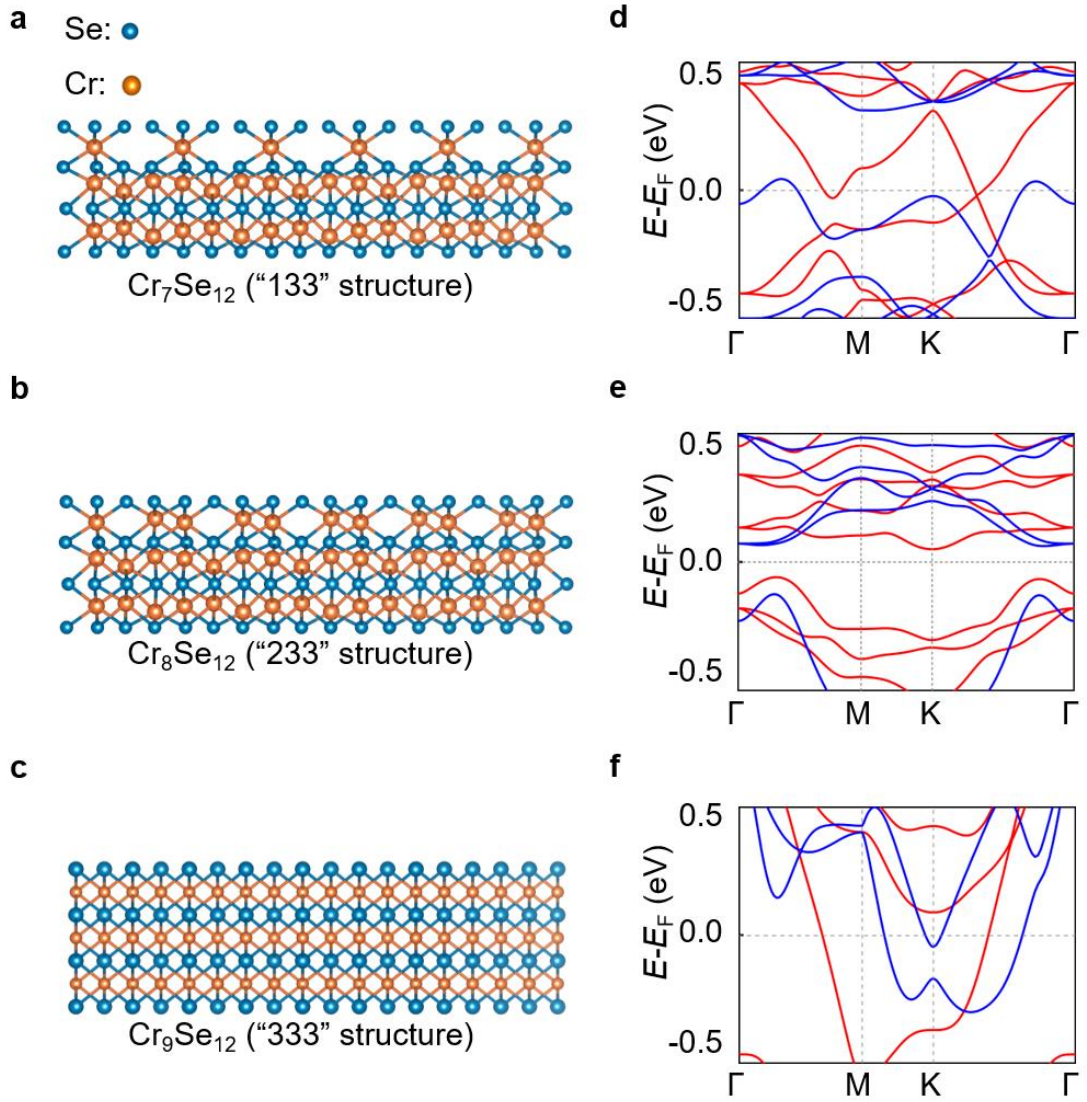
Supplementary Figure 2. The evolution of the STM topography with respect to the deposition time. (a, b, c) Large-scale STM topographies of the 2D chromium selenide samples with deposition time of (a) 20 minutes, (b) 40 minutes, and (c) 60 minutes. Scale bars: 140 nm. The coverage ratio of 2D Cr_8Se_{12} to 2D Cr_2Se_3 increases with the increased deposition time.

Part 2. Cross-sectional STEM characterization of 2D Cr_8Se_{12}

To further validate the 7-atomic-layer crystal structure of 2D Cr_8Se_{12}, we conducted the cross-sectional STEM characterization. To facilitate the focused ion beam lithography, we grew the sample on the epitaxial graphene-terminated n-type 4H-SiC(0001) under the same condition. We noticed that 2D Cr_8Se_{12} can also be obtained on this substrate, as shown in Supplementary Figure 3a. The as-grown sample was then capped with ~ 25 nm of Se by deposition of Se at room temperature to protect the sample from surface contamination. From the cross-sectional STEM image displayed in Supplementary Figure 3b, we can identify 4 atomic layers of Se with 3 atomic layers of Cr in between, validating the proposed 7-atomic-layer crystal structure of 2D Cr_8Se_{12}.



Supplementary Figure 3. Cross-sectional STEM characterization on the 2D Cr_8Se_{12}. (a) STM topography of 2D Cr_2Se_3 grown on the epitaxial graphene-terminated 4H-SiC indicating that both the 2D Cr_8Se_{12} and 2D Cr_2Se_3 flakes are obtainable on this substrate. It is notable that the 2D Cr_8Se_{12} and 2D Cr_2Se_3 flakes may cross the SiC steps covered by uniform epitaxial graphene. Scale bar: 20 nm. (b) Cross-sectional STEM image of the 2D Cr_8Se_{12} grown on the epitaxial graphene-terminated 4H-SiC. Scale bar: 1 nm. The side-view schematic crystal structure of the 7-atomic-layer Cr_2Se_3 displayed in Fig. 2h is superimposed here for clarity.



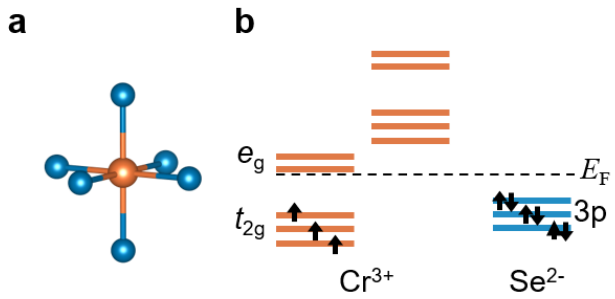
Supplementary Figure 4. Three candidate crystal structures for the thicker flake with 7-atomic-layer thickness and their DFT band structures. (a, b, c) Side-view schematic models of three candidate crystal structures with (a) one, (b) two, and (c) three Cr atoms in the Cr adlayer for each unit cell. (d, e, f) DFT band structures for the crystal structures displayed in panel a, b and c, respectively. It is notable that only the "233" structure with the stoichiometric ratio of Cr/Se = 8/12 exhibits a bandgap around the Fermi level.

Supplementary Table 1. Comparison on experimentally reported chromium selenide compounds with different crystal structures and lattice constants.

Materials	In-plane lattice constant *a* (nm)	In-plane lattice constant *b* (nm)	In-plane angle *θ* (°)	Sample form	Supplementary Reference number
CrSe	0.3674 ± 0.0004	0.3674 ± 0.0004	120	bulk	2
$CrSe_2$	0.3399	0.3399	120	bulk	3
$CrSe_2$	0.361	0.617	90	bulk	4
Cr_3Se_4	0.3598(1)	0.6270(1)	90	bulk	5
Cr_5Se_8	0.7148 ± 0.0004	1.2353 ± 0.0004	91.06 ± 0.01	bulk	6
Cr_7Se_8	0.749	0.752	119.74	bulk	7
Cr_2Se_3	0.625 ± 0.003	0.625 ± 0.003	120	bulk	8
Cr_2Se_3	0.370 ± 0.003	0.370 ± 0.003	120	2D	This work
Cr_8Se_{12}	0.634 ± 0.004	0.634 ± 0.004	120	2D	This work

Part 3. Theoretical analysis on the semiconducting nature.

The semiconducting nature of 2D Cr_8Se_{12} can be understood through an analysis on its crystal structure and the orbital occupation configuration. Being in the octahedral coordination environment, the Cr^{3+} cations experience an octahedral crystal field, causing a splitting of the 3*d* orbitals into two e_g orbitals and three t_{2g} orbitals for each spin channel. As each Cr^{3+} cation has only three valence electrons, the minority spin orbitals remain unoccupied, resulting in a wide bandgap between the majority spin e_g and t_{2g} orbitals (See Supplementary Figure 5).



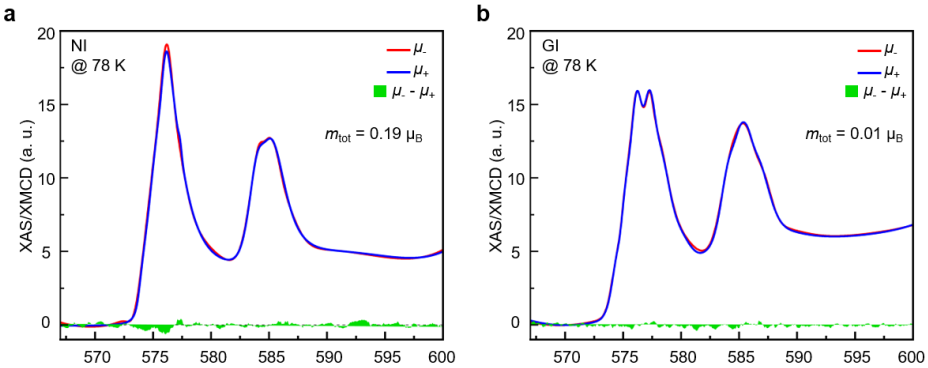
Supplementary Figure 5. Theoretical analysis on the semiconducting nature of 2D Cr_8Se_{12}. (a) Schematic demonstration of the octahedral crystal field. (b) Schematic representation of the crystal field induced energy splitting to account for the semiconducting nature.

Part 4. Discussion on magnetic configurations.

Owing to the stoichiometric ratio of Cr/Se = 8/12 in 2D Cr_8Se_{12}, the average valence state of the Cr cations is +3 with the electron configuration of [Ar] $3d^3$. The presence of three valence electrons in the Cr^{3+} cation results in an unfilled shell, leading to an approximate magnetic moment of 3 μ_B. Consequently, 2D Cr_8Se_{12} is expected to exhibit distinct magnetic properties. To investigate the magnetic properties, the x-ray absorption spectroscopy (XAS) was performed at the Cr $L_{2,3}$ edges under magnetic fields of $B = \pm 1.5$ T, with the x-ray magnetic circular dichroism (XMCD) signal obtained as the difference between the XAS pairs with opposite magnetic fields. The XMCD characterizations were conducted on the chromium selenide sample with sub-monolayer coverage, both in the normal incidence (NI) and grazing incidence (GI) configurations, to examine the magnetic behavior along the in-plane and out-of-plane directions, respectively.

With the Cr^{3+} cation having a hole number $N^h = 7$, the XMCD sum rules allowed for an element-selective measurement of the total magnetic moment ($m_{tot} = m_L + m_S$) of the Cr element. As shown in Supplementary Figure 6a and Supplementary Figure 6b, the XMCD characterizations performed at 78 K reveal total magnetic moments of approximately 0.42 μ_B and 0.01 μ_B for the NI and GI configurations, respectively. The larger total magnetic moment along the out-of-plane direction compared to the in-plane direction indicates the out-of-plane magnetic anisotropy. We notice that the out-of-plane total magnetic moment of 0.42 μ_B is significantly lower than the expected value

of ~ 3 μ_B for the ferromagnetic configuration. Consequently, it can be inferred that the magnetic ground state of the 2D Cr_8Se_{12} might be ferrimagnetic with interlayer antiferromagnetic coupling.



Supplementary Figure 6. Out-of-plane anisotropy and ferrimagnetic configuration. (a, b) XAS and XMCD spectra of 2D chromium selenide/graphite measured at 78 K in the NI (a) and GI (b) configuration, respectively.

Furthermore, as shown in Supplementary Table 2 and Supplementary Table 3, we also performed DFT calculation on the total energies of 2D Cr_8Se_{12} and 2D Cr_2Se_3 with different magnetic configurations. It is notable that the ferrimagnetic configuration for 2D Cr_8Se_{12} and the A-type antiferromagnetic configuration for 2D Cr_2Se_3 exhibit the lowest energies. Therefore, we calculated the band structures of 2D Cr_8Se_{12} and 2D Cr_2Se_3 with ferrimagnetic and interlayer antiferromagnetic configurations, respectively, and reproduced the measured ARPES dispersion shown in Fig. 3b and 4a.

Supplementary Table 2. DFT calculated total energy of 2D Cr_8Se_{12} with different magnetic configurations.

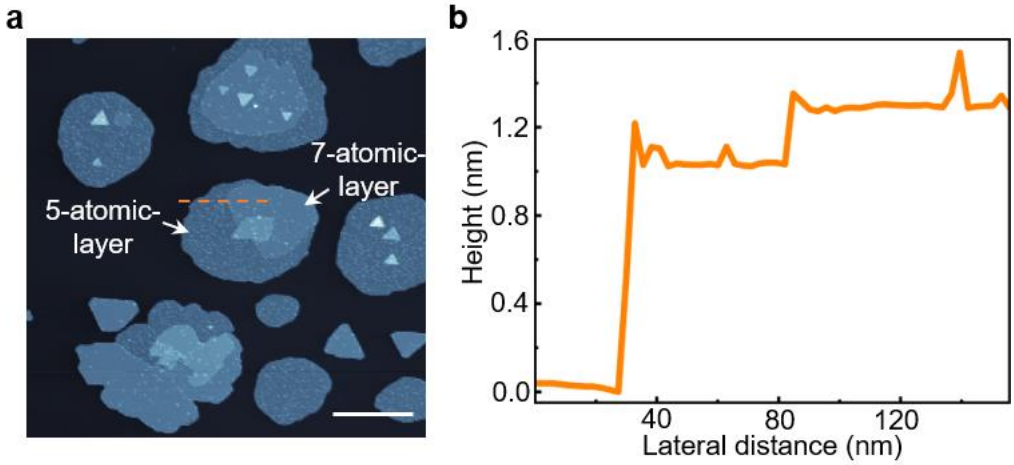
Magnetic order	ΔE (meV/atom)
nonmagnetic	186.6
ferromagnetic	68.0
ferrimagnetic	0

Supplementary Table 3. DFT calculated total energy of 2D Cr_2Se_3 with different magnetic configurations.

Magnetic order	ΔE (meV)
nonmagnetic	398.8
ferromagnetic	1.3
A-type antiferromagnetic	0

Part 5. Environmental stability of 2D Cr_8Se_{12}

To examine the environmental stability of 2D Cr_8Se_{12}, we exposed the as-grown sample in air at room temperature for 70 minutes and reloaded it into the STM system for characterization. The air-exposed sample was annealed in ultra-high vacuum at around 300 °C for 30 minutes before the STM characterization. As shown in Supplementary Figure 7a and Supplementary Figure 7b, we notice that the morphology of both 2D Cr_8Se_{12} and 2D Cr_2Se_3 remains intact except for several adsorbates introduced by the air exposure.



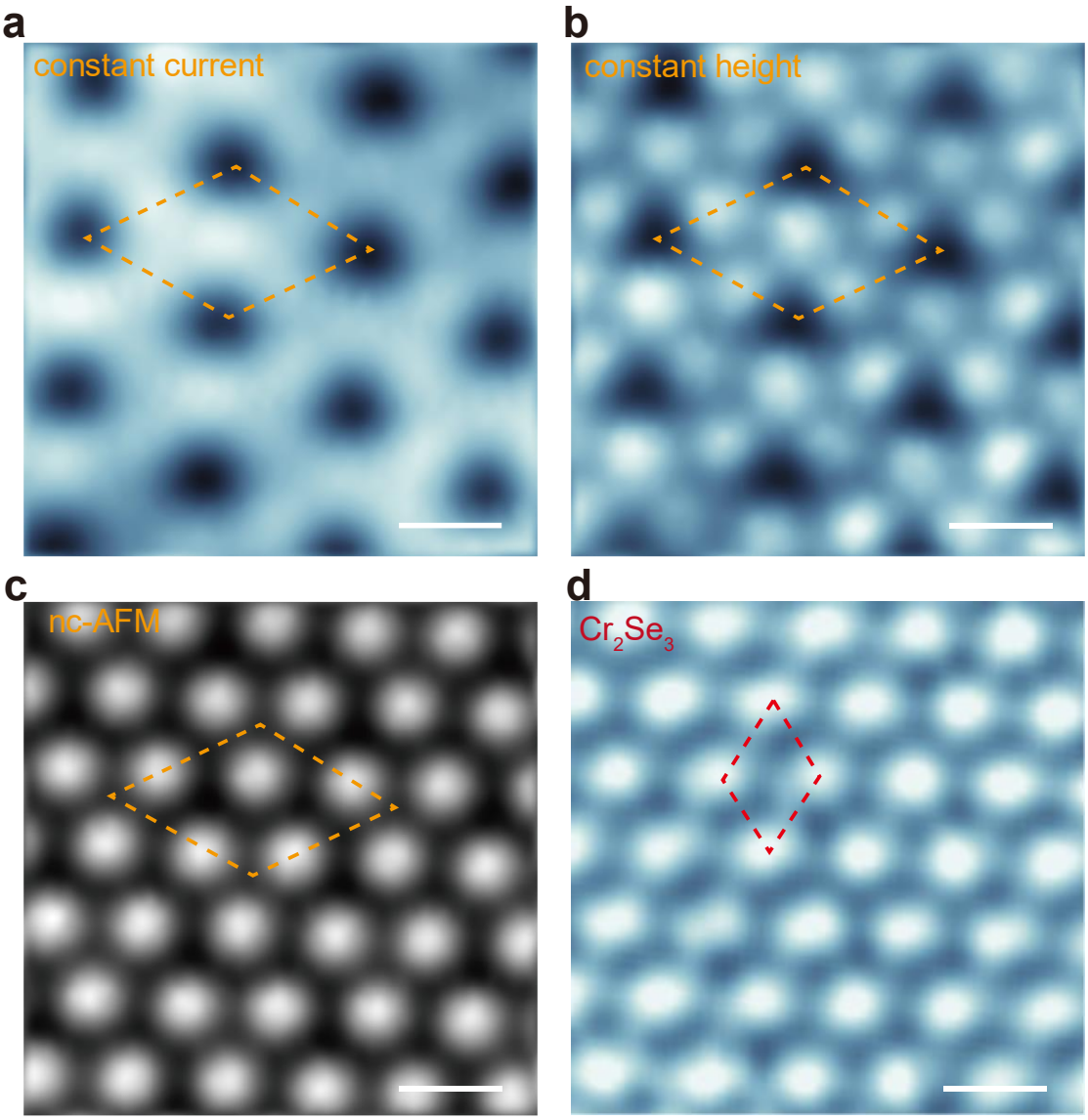
Supplementary Figure 7. Environmental stability of 2D Cr_8Se_{12}. (a) STM topography of the chromium selenide/graphite sample after exposure in air for 70 minutes. Scale bar: 140 nm. (b) STM line profile along the orange dashed line in panel a.

Part 6. Details about comparing the DFT band structures with the measured ARPES dispersion.

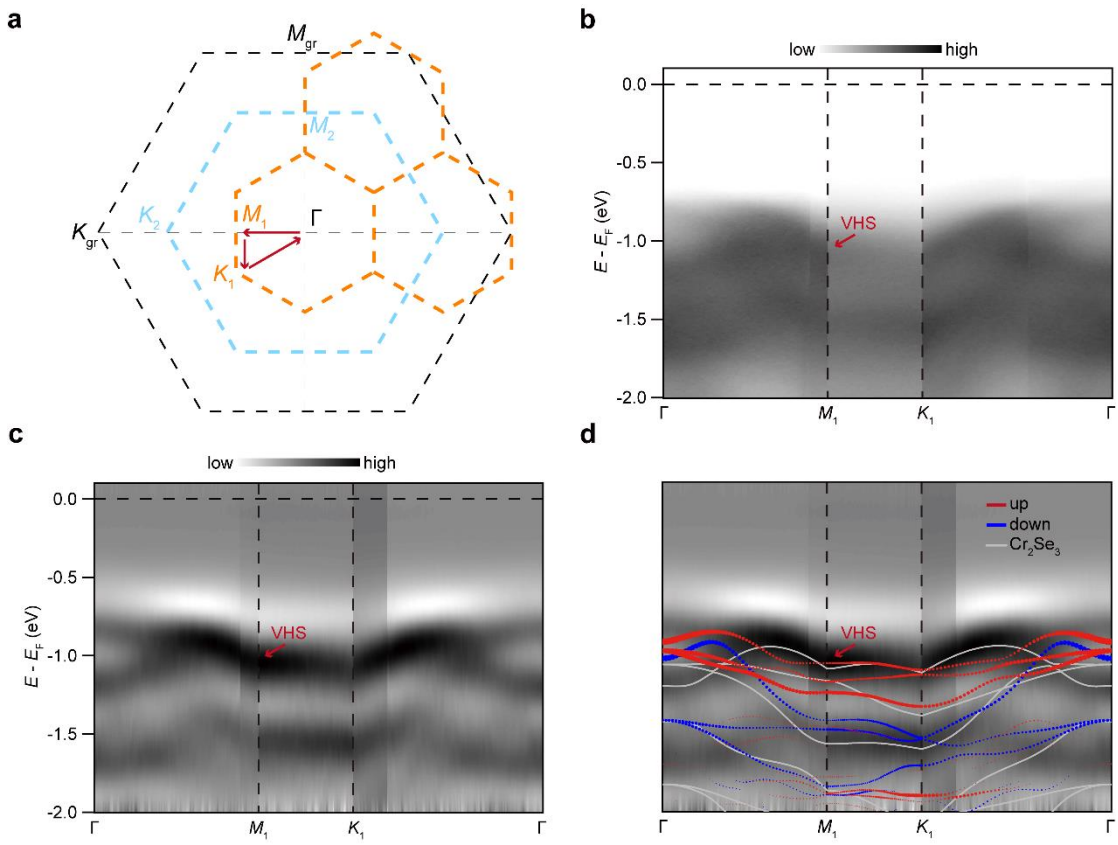
We conducted dI/dV measurements on the electronic properties of both 2D Cr_8Se_{12} (Fig. 3e) and 2D Cr_2Se_3 (Supplementary Figure 8a) and determined that both two phases exhibit a bandgap below the Fermi level. The DFT calculations were also performed to examine the band structures of these two phases (Supplementary Figure 4e for 2D Cr_8Se_{12} and Supplementary Figure 8b for 2D Cr_2Se_3). The semiconducting properties revealed in the DFT band structures are consistent with the experimental observations. We notice that the DFT gap sizes are smaller than the experimental values. This is probably due to the underestimation of the band gap with the PBE functional in DFT calculations.[9] We also experimentally observed the Fermi level appearing near the conduction band minimum (CBM) rather than inside the bandgap, which might originate from the higher work functions of 2D Cr_8Se_{12} and 2D Cr_2Se_3 than the graphite substrate. Consequently, to compare the DFT valence band structures with the ARPES dispersion, we shifted the theoretical Fermi levels upwards to align the DFT valence band maximum (VBM) with the measured VBM in dI/dV spectra.

Part 7. Detailed structural characterization of 2D Cr_8Se_{12}

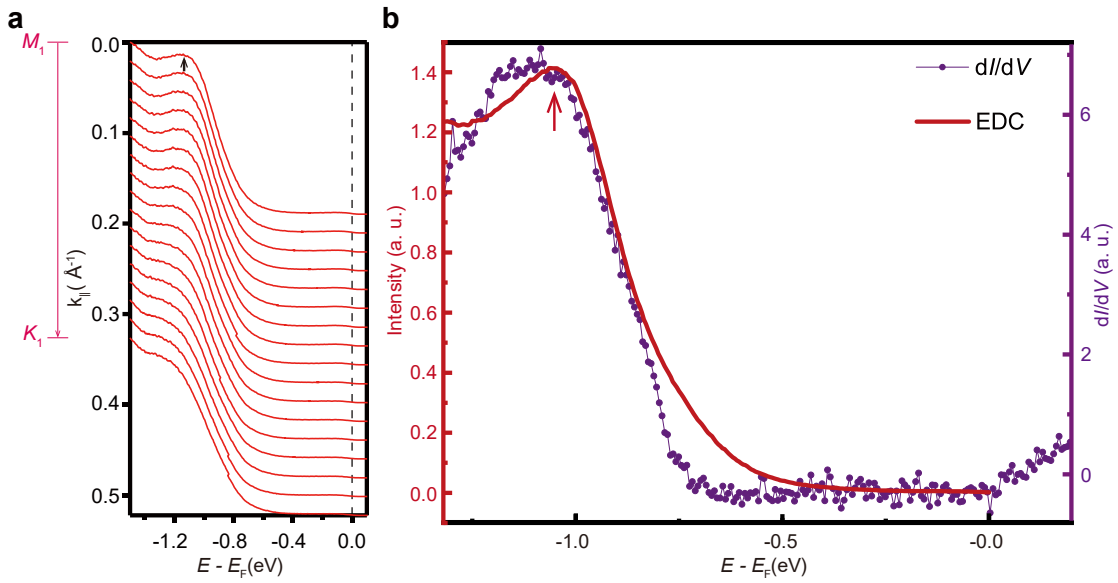
To demonstrate the √3×√3 reconstructed crystal structure of 2D Cr_8Se_{12}, we performed both constant current and constant height STM measurements and compared the results with the constant current STM images of 2D Cr_2Se_3. As demonstrated in Supplementary Figure 8a and Supplementary Figure 8b, both the constant current and constant height STM images exhibit the √3×√3 reconstruction. However, the constant current STM images of 2D Cr_2Se_3 in Supplementary Figure 8d and Supplementary Figure 1a suggest that there is no such reconstruction in Cr_2Se_3. Therefore, by comparing their constant current topographies in Supplementary Figure 8a and Supplementary Figure 8d, we could clearly distinguish Cr_8Se_{12} and Cr_2Se_3. In addition, the √3×√3 reconstruction of 2D Cr_8Se_{12} can be further confirmed by comparison of the typical STM constant current topography with the nc-AFM image (Supplementary Figure 8c).



Supplementary Figure 8. Determination of the √3×√3 R30° reconstruction on 2D Cr_8Se_{12}. (a, b) Constant current (a) and constant height (b) STM topography of 2D Cr_8Se_{12} taken at V_s = 20 meV. (c) Nc-AFM image of 2D Cr_8Se_{12} taken at the same scanning window as panel (a) and panel (b). (d) Constant current STM topography of 2D Cr_2Se_3 taken at V_s = -100 meV. The unit cells of 2D Cr_8Se_{12} in panel (a-c) are indicated by orange dashed rhombuses, and the unit cell of 2D Cr_2Se_3 is indicated by red dashed rhombus. Scale bars in panel (a-d) is 0.4 nm.



Supplementary Figure 9. Direct comparison between ARPES and DFT band structures within the 2D Cr_8Se_{12} BZ. (a) The geometric relation between the BZs of 2D Cr_8Se_{12} (orange), 2D Cr_2Se_3 (blue), and the graphite substrate (black), adapted from Fig. 3a. The red arrows indicate the path along which the ARPES cuts in panel (b) are taken. (b, c) Raw ARPES (a) and second-derivative ARPES (b) cuts of *in-situ* grown sample within the 2D Cr_8Se_{12} Brillouin zone. (d) Aligned DFT bands of 2D Cr_8Se_{12} (blue and red) and 2D Cr_2Se_3 (white) for comparison with the ARPES data.



Supplementary Figure 10. (a) The ARPES energy dispersion curves (EDCs) along the M_1-K_1 direction which indicate the existence of VHS at ~ -1.10 eV. (b) Comparison of the integrated EDC curve with the dI/dV curve.

Part 8. More details on the tight binding modeling

To examine the extent to which the tight binding modeling reflects the realistic situation, we performed detailed analysis on the tight binding results.

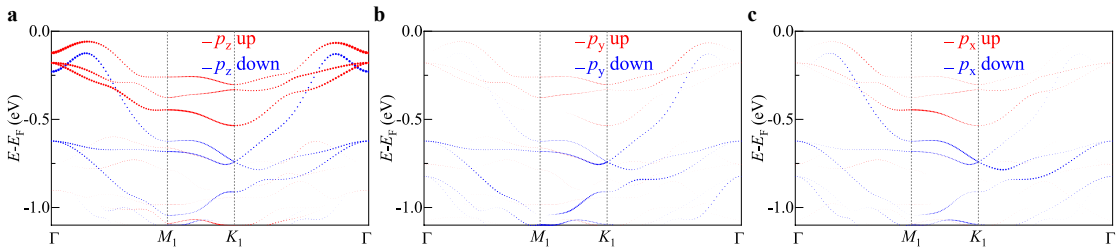
Firstly, to evaluate the contribution of p_x, p_y mixing, we analyzed the projected band structures of 2D Cr_8Se_{12} with respect to the top layer Se p_x, p_y and p_z orbitals. The results in Supplementary Figure 11 demonstrate that the p_z orbitals contribute significantly more to the band structure than the Se p_x or p_y orbitals, especially for the spin up channel near the VBM. The predominant contribution of Se p_z orbitals indicates that the single Se p_z-orbital approximation is reasonable.

To further confirm the validity of the single orbital approximation, we calculated the Wannier bands using Se p_z orbitals as the basis and compared them with the DFT bands. As depicted in Supplementary Figure 12b, the Wannier bands perfectly reproduce the DFT bands, confirming the dominant role of the Se p_z orbitals in the energy bands near the VBM.

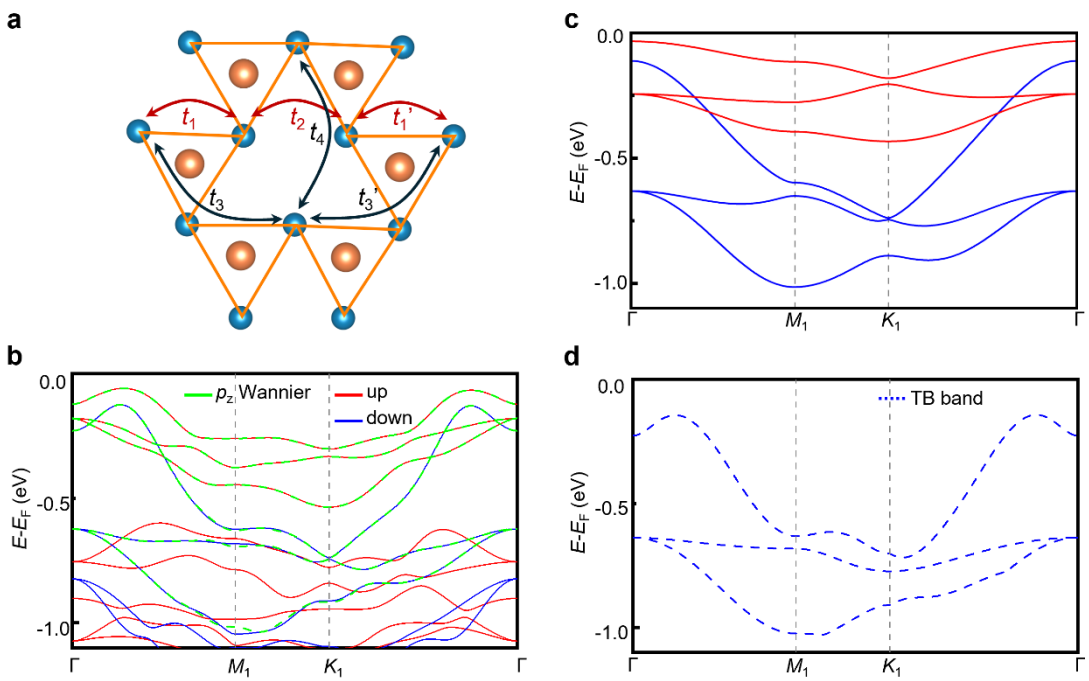
Considering the dominant contributions of Se p_z orbitals to the energy bands near the VBM, we included the next-nearest-neighbor (nnn) hopping terms to improve the TB model. The band structure with both nearest-neighbor (nn) and nnn hopping is shown in Supplementary Figure 12c, with the hopping parameters defined and listed in Supplementary Figure 12a and Supplementary Table 5.

Regarding the residual discrepancies between the DFT and TB bands near the VBM, we deduce that they may be due to the inevitable orbital mixing between the Se p_z

orbitals and other electronic states. In Supplementary Figure 11, we observe that p_x and p_y orbital mixing plays a more significant role in the spin-down channel than in the spin-up channel. To improve the TB model, we included the orbital mixing of the p_z orbitals with the p_x and p_y orbitals in the spin-down channel using the Slater-Koster TB method.[10] The Slater-Koster TB parameters are obtained by fitting the DFT bands with the TBStudio software,[11] with the parameters displayed in Supplementary Table 5. As shown in Supplementary Figure 12d, the inclusion of orbital mixing significantly enhances the similarity between the TB and DFT bands. This improvement indicates that considering orbital mixing helps resolve the discrepancy between the TB band without orbital mixing and the DFT band.



Supplementary Figure 11. Projected band structure of 2D Cr_8Se_{12} with respect to the top layer Se (a) p_z (b) p_y and (c) p_x orbitals, with the spin-up and spin-down contributions indicated by red and blue dots, respectively.



Supplementary Figure 12. (a) TB model of the CT lattice on 2D Cr_8Se_{12} with three types of nn hopping parameters denoted by t_1, t_1', t_2 and three types of nnn hopping parameters denoted by t_3, t_3', t_4. (b) Comparison of the DFT bands (solid lines) and the Wannier bands with Se-p_z characteristics (green dashed lines). (c) TB bands of the CT lattice on 2D Cr_8Se_{12} with both the nn and the nnn hopping. (d) TB band of the CT lattice considering nn, nnn hopping and the p_x, p_y, p_z orbital mixing.

Supplementary Table 4. Tight binding parameters used for the simulation in Fig. 4c.

spin component	t_1	t_1'	t_2
up	0.054	0.046	-0.030
down	0.151	0.149	-0.045

Supplementary Table 5. Tight binding parameters used for the simulation in Supplementary Figure 12c.

spin	t_1	t_1'	t_2	t_3	t_3'	t_4
up	0.054	0.046	-0.030	0.016	0.014	0.003
down	0.079	0.069	0.025	0.064	-0.017	0.064

Supplementary Table 6. Slater-Koster TB parameters used for the simulation in Supplementary Figure 12d.

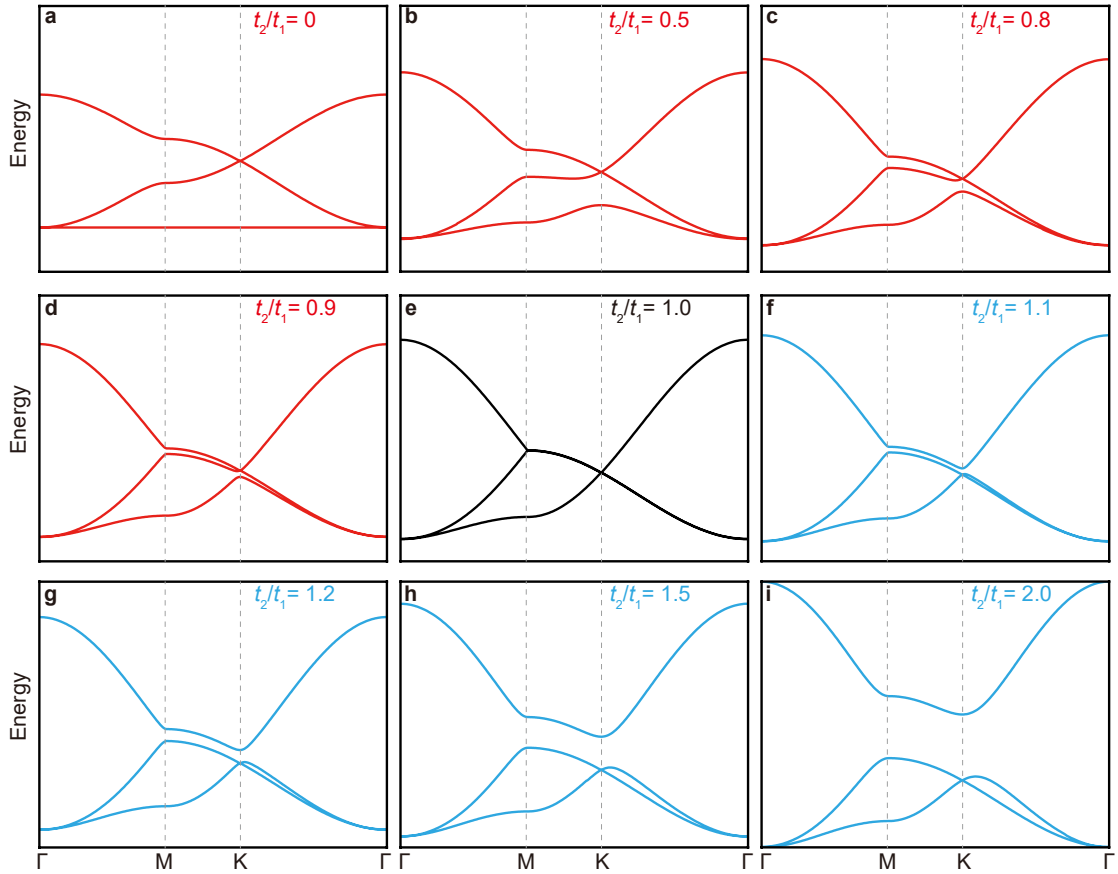
nn hopping	bond 1	$V_{pp\sigma}$	1.248
		$V_{pp\pi}$	0.130
	bond 1’	$V_{pp\sigma}$	-1.595
		$V_{pp\pi}$	-0.761
	bond 2	$V_{pp\sigma}$	-0.039
		$V_{pp\pi}$	0.034
nnn hopping	bond 3	$V_{pp\sigma}$	0.565
		$V_{pp\pi}$	0.019
	bond 3’	$V_{pp\sigma}$	0.231
		$V_{pp\pi}$	0.172
	bond 4	$V_{pp\sigma}$	0.048
		$V_{pp\pi}$	-0.016

Part 9. The evolution of CT lattice band structures with respect to t_2/t_1

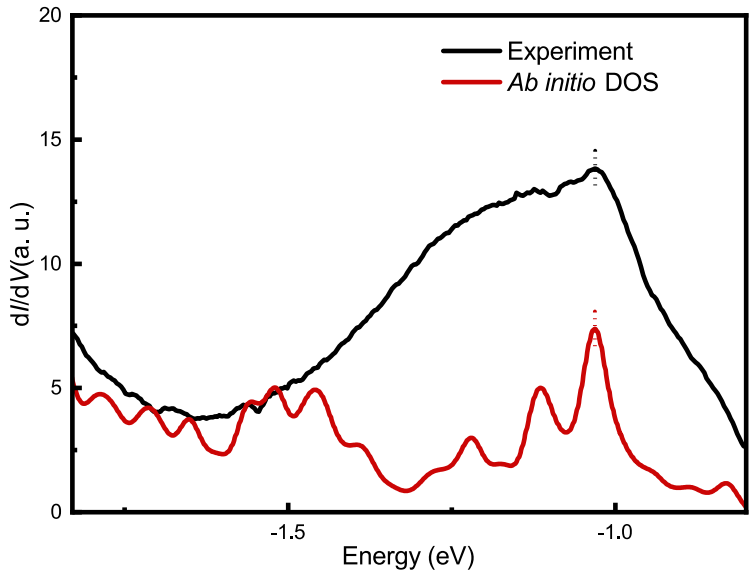
To elaborate the fundamental property of CT lattice, we investigated the evolution of its band structures for t_2/t_1 varying from 0 to 2, especially the one for t_2/t_1 =1.

As demonstrated in Supplementary Figure 13a-13d, for $t_2/t_1 < 1$, there is a two-fold band crossing at the K point formed by the top two bands, with the bottom band being separated. However, for $t_2/t_1 > 1$, as shown in Supplementary Figure 13e-13j, it is the top band that is separated, with the two-fold band crossing at the K point formed by the bottom two bands. At the critical point, i.e., $t_2/t_1 = 1$, as shown in Supplementary Figure 13e, the upper two bands are doubly degenerate all the way along M-K, making the crossing point at the K point triply degenerate.

In the case of $t_2/t_1 = 1$, the TB model depicts the band structure of a triangular lattice with a $\sqrt{3}\times\sqrt{3}$ supercell, resulting a triply degenerate point at the *K* point. It is important to note that due to the 2/3 occupied Cr layer, the $\sqrt{3}\times\sqrt{3}$ supercell is the primitive unit cell of 2D Cr_8Se_{12}. For $t_2/t_1 < 1$, the second and third bands intersect at the *K* point. For $t_2/t_1 >1$, there is a band inversion between the second and third bands, leading to a doubly degenerate point between the first and second bands. Therefore, in terms of t_2/t_1 phase diagram, there is a topological phase transition occurring.

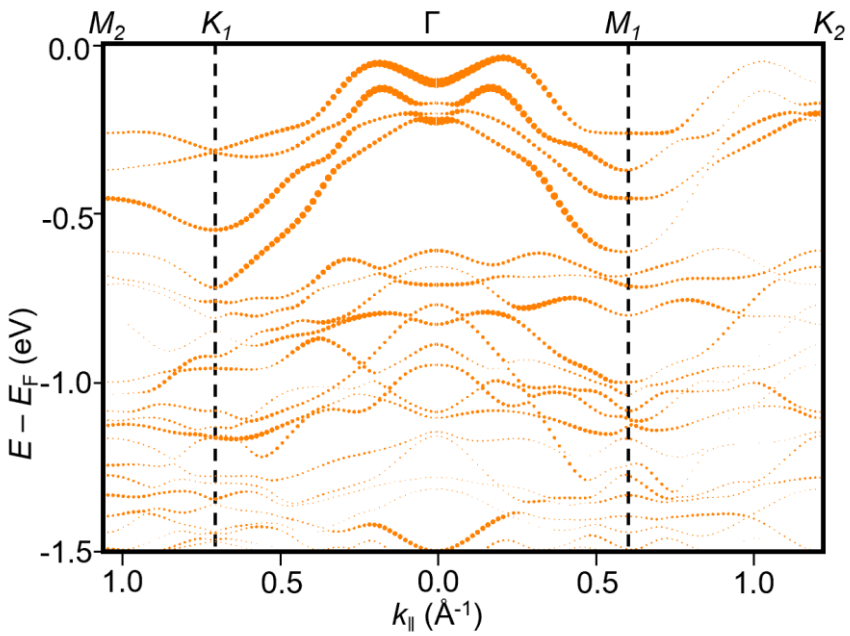


Supplementary Figure 13. Band evolution of CT lattice. (a-d) Band structures of CT lattice with $t_2/t_1 < 1$. (e) Band structure of CT lattice with $t_2/t_1 = 1$, which corresponds to the triangular lattice. (f-i) Band structures of CT lattice with $t_2/t_1 > 1$.

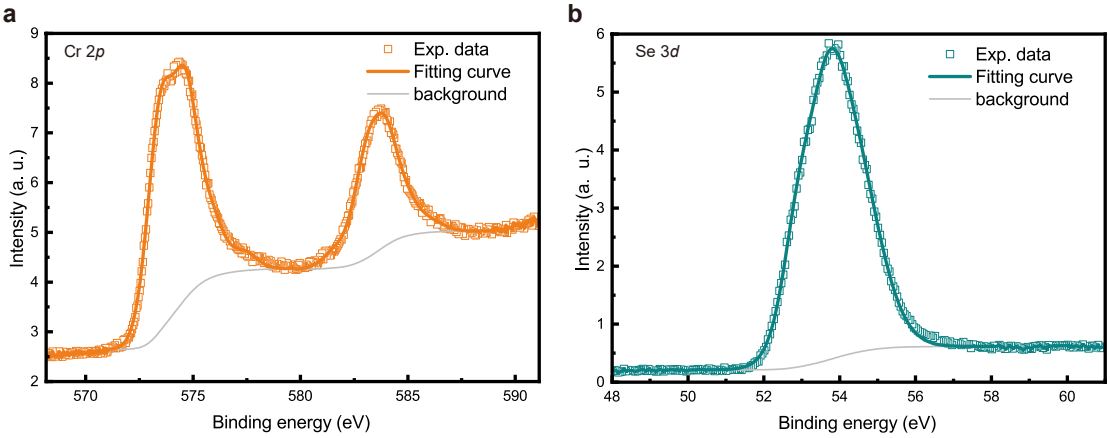


Supplementary Figure 14. Direct comparison of d*I*/d*V* curve and the ab initio calculated density of states (DOS). The pronounced peaks originating from the VHS are denoted by vertical dashed lines.

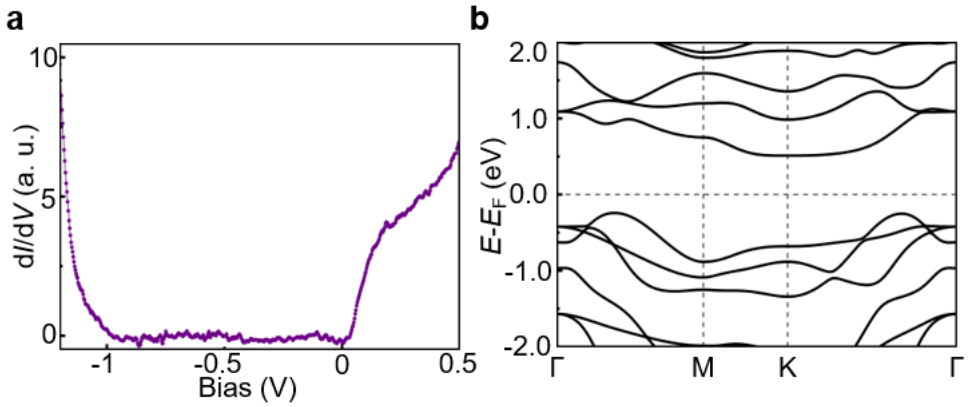
The slight discrepancy between the *Ab initio* DOS and the measured d*I*/d*V* might be due to the tunneling matrix elements and the instrumental broadening. Firstly, according to Bardeen tunneling theory, the tunneling current (*I*) is proportional to the integral of the product of LDOS and $|M|^2$, where $|M|$ represents the tunneling matrix element. This matrix element can vary for different states in a band, leading to a bias-dependent tunneling transmission term $|M|^2$. Additionally, the energy broadening, caused by the thermal effects and the modulation voltage of the lock-in amplifier used in d*I*/d*V* measurements, could also significantly contribute to the differences between the calculated LDOS and the measured d*I*/d*V* curve.



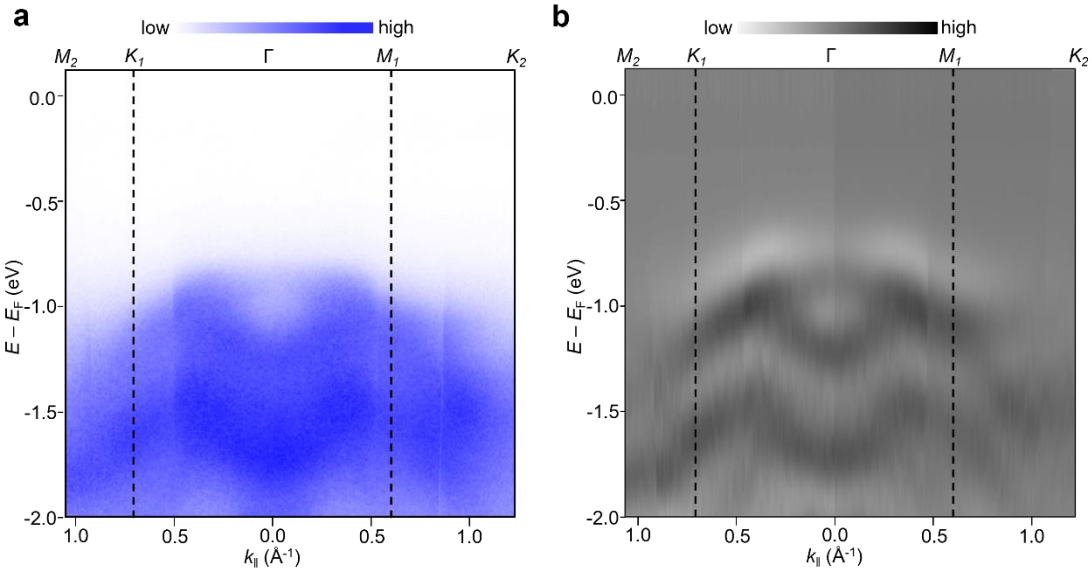
Supplementary Figure 15. Unfolded DFT band structures of 2D Cr_8Se_{12}, taken along the high-symmetry lines presented in the ARPES results in Fig. 3b.



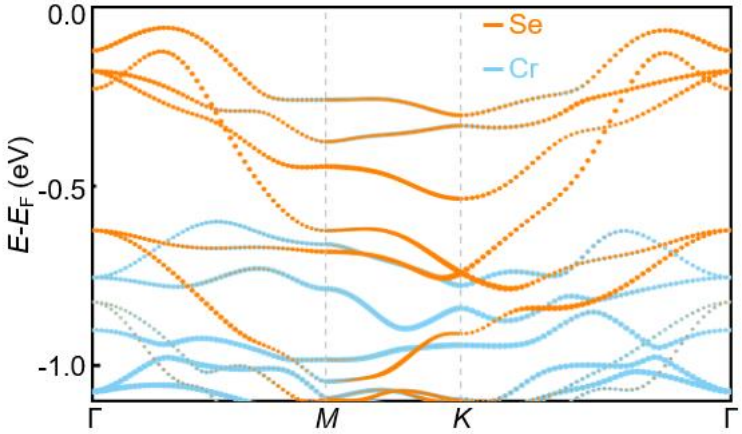
Supplementary Figure 16. (a, b) XPS core level spectra of (a) Cr 2*p* and (b) Se 3*d* for determination of the stoichiometry of the as-grown sample. With quantitative XPS analysis, we determined the stoichiometric ratio of Cr : Se = 8 : 13, which is comparable to the theoretical value of Cr : Se = 8 : 12.



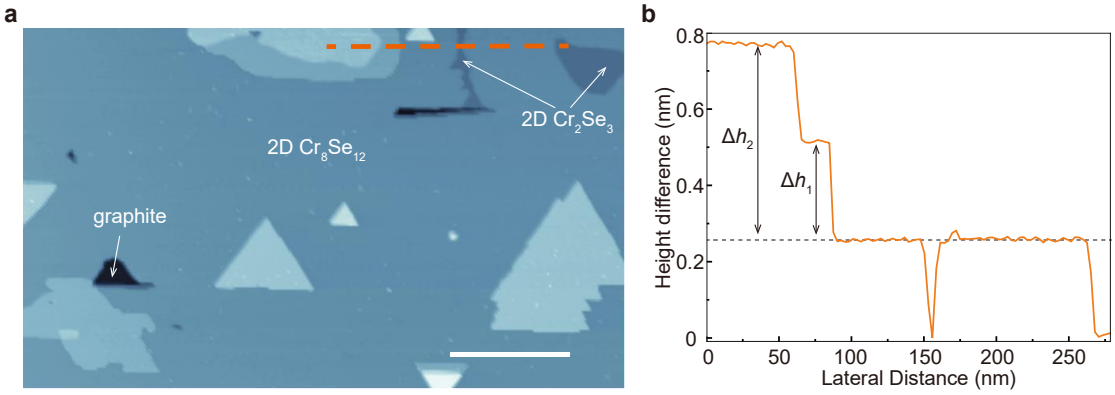
Supplementary Figure 17. Electronic properties of 2D Cr_2Se_3. (a) d*I*/d*V* spectrum acquired on 2D Cr_2Se_3. (b) DFT band structures of freestanding 2D Cr_2Se_3.



Supplementary Figure 18. ARPES dispersions of chromium selenide. ARPES (a) and second-derivative ARPES (b) dispersions of as-grown chromium selenide on graphite, presenting ARPES data without overlaid DFT band structures. Panel b is adapted from Fig. 3b to enhance the clarity of the ARPES data.



Supplementary Figure 19. Projected electronic band structure of 2D Cr_8Se_{12} highlighting topmost Se p_z orbital contributions. The band structure features significant contributions from the topmost Se p_z orbitals in the electronic states around the VBM, as indicated by the orange dots.



Supplementary Figure 20. STM investigation of chromium selenide sample with coverage of larger than one Cr_8Se_{12} unit cell. (a) Large-scale STM topography of an area with thicker chromium selenide island. Scale bar, 140 nm. (b) Line profile along the orange dashed line in panel a. The STM topographic height differences between the thicker islands and 2D Cr_8Se_{12} are $\Delta h_1 \sim 0.29$ nm and $\Delta h_2 \sim 0.57$ nm, respectively.

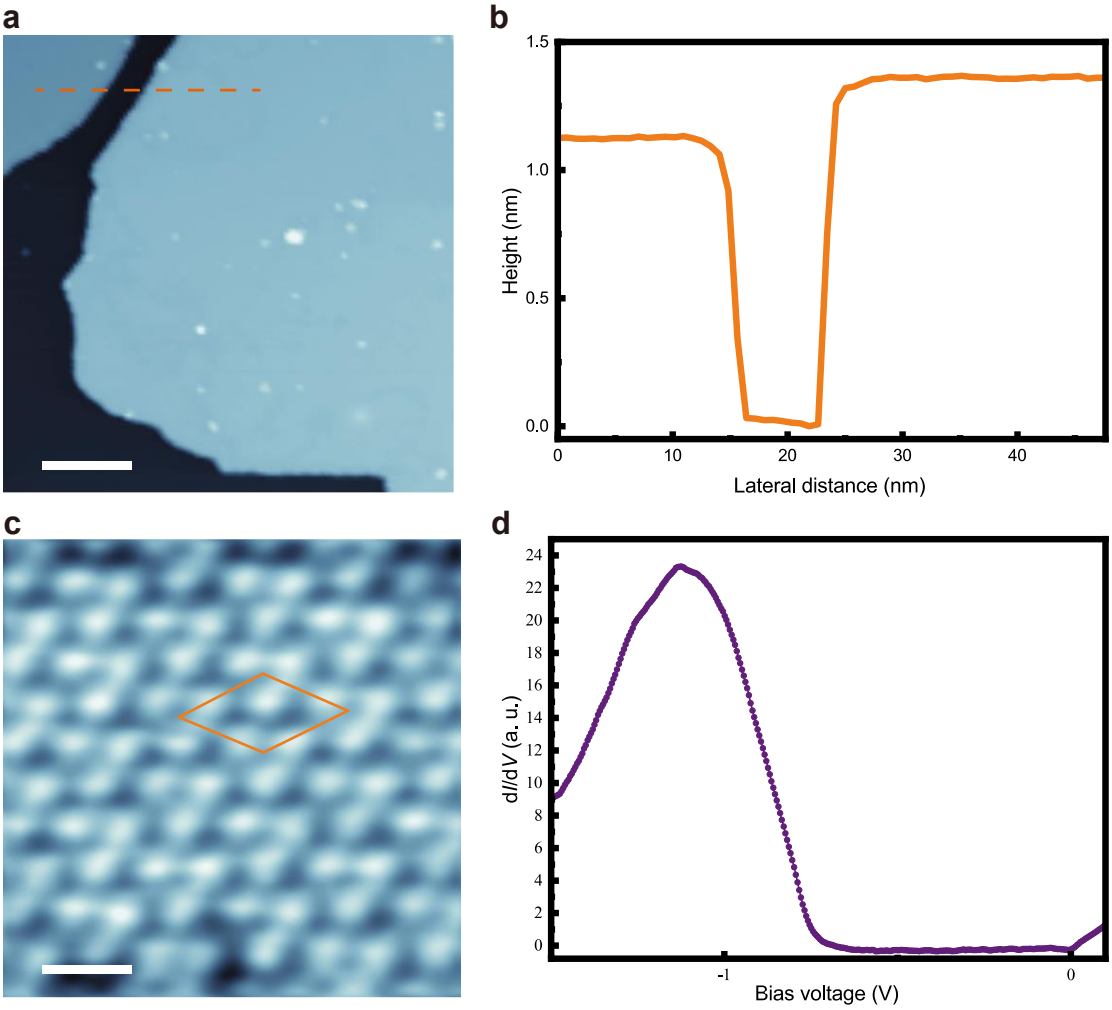
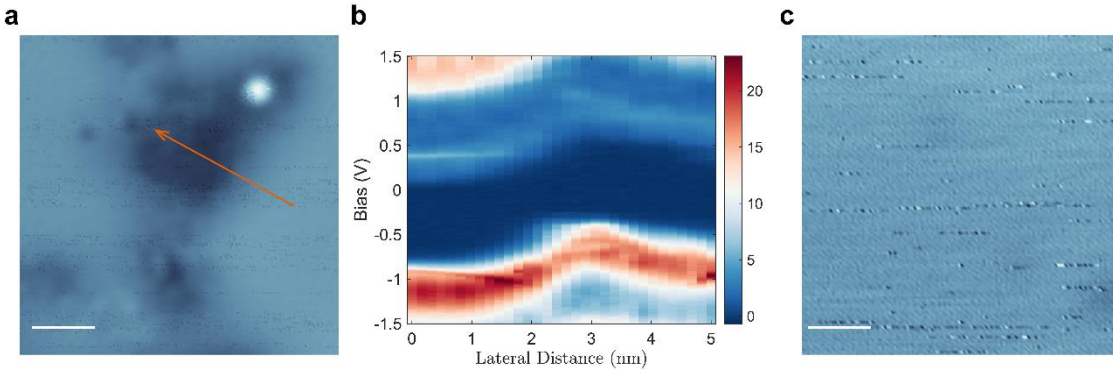
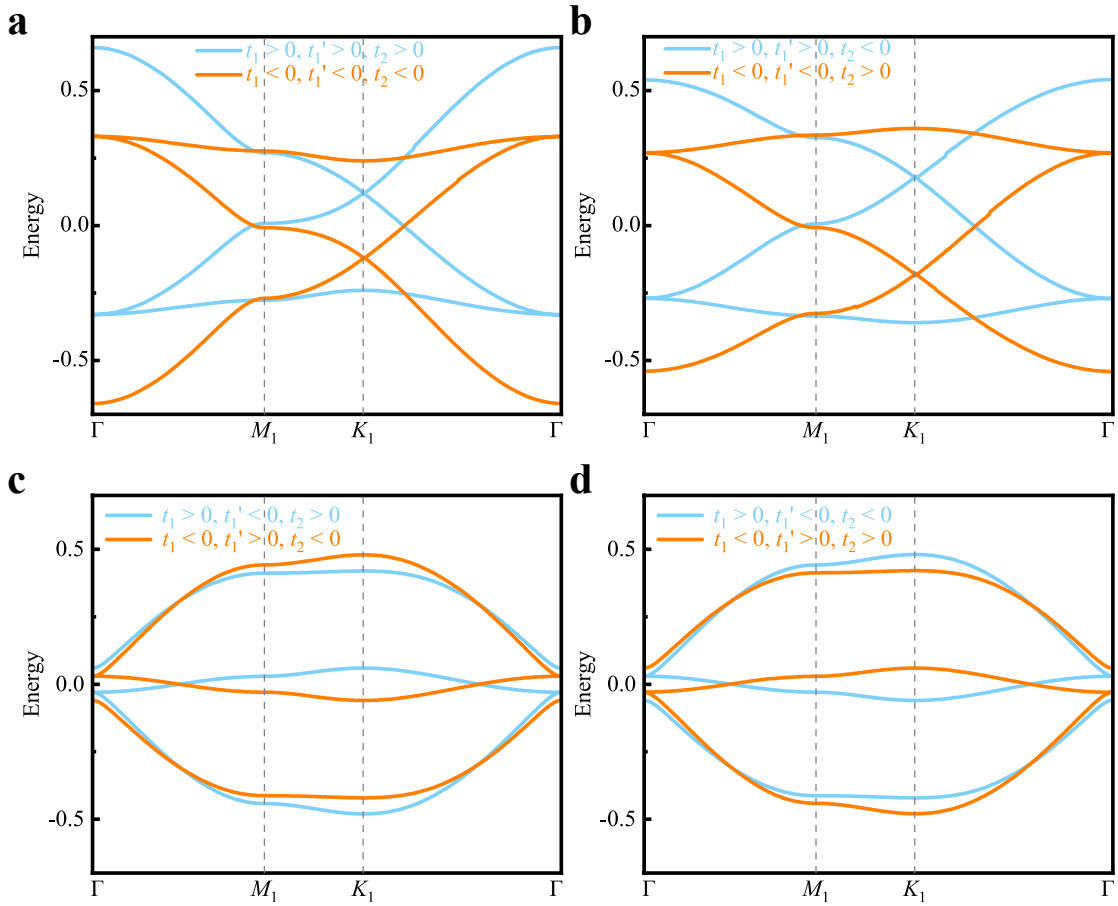


Fig S21. Crystal structure and STS characterizations of decapped 2D Cr_8Se_{12} sample. (a) Large-scale STM topography of the decapped chromium selenide sample. The orange dashed line crosses both a 2D Cr_2Se_3 and a 2D Cr_8Se_{12} island. (b) STM line profile along the orange dashed line in panel a, demonstrating that the thickness of the decapped sample is consistent with that of the as-grown sample. (c, d) Atomic-resolution constant current STM image (c) and dI/dV spectrum (d) of the decapped 2D Cr_8Se_{12} sample, revealing that the in-plane constant and electronic properties of the decapped sample match those of the as-grown sample.



Supplementary Figure 22. Comparison of the sample quality between 2D Cr_2Se_3 and 2D Cr_8Se_{12}. (a) Typical STM topography of 2D Cr_8Se_{12} with domain walls. (b) STS line spectroscopy taken along the orange arrow in panel a, which crosses both the normal domain region and the domain wall region. (c) Typical STM topography of 2D Cr_2Se_3 without domain walls. Scale bars in panel a and panel c, 2 nm.



Supplementary Figure 23. (a, b) The CT lattice band structures with t_1 and t_1' having the same sign. (a) t_1 and t_2 have the same sign. The blue (orange) bands represent the band structures in the case of $t_1 = t_1' = 0.15$ and $t_2 = 0.2\ t_1 = 0.03$ ($t_1 = t_1' = -0.15$ and $t_2 = 0.2\ t_1 = -0.03$). (b) t_1 and t_2 have the opposite signs. The blue (orange) bands represent the band structures in the case of $t_1 = t_1' = 0.15$ and $t_2 = -0.2\ t_1 = -0.03$ ($t_1 = t_1' = -0.15$ and $t_2 = -0.2\ t_1 = 0.03$). (c, d) The CT lattice band structures with t_1 and t_1' having the opposite sign. (c) t_1 and t_2 have the same sign. The blue (orange) bands represent the band structures in the case of $t_1 = 0.15$, $t_1' = -0.15$ and $t_2 = 0.2\ t_1 = 0.03$ ($t_1 = -0.15$, $t_1' = 0.15$ and $t_2 = 0.2\ t_1 = -0.03$). (d) t_1 and t_2 have the opposite signs. The blue (orange) bands represent the band structures in the case of $t_1 = 0.15$, $t_1' = -0.15$ and $t_2 = -0.2\ t_1 = -0.03$ ($t_1 = -0.15$, $t_1' = 0.15$ and $t_2 = -0.2\ t_1 = 0.03$).

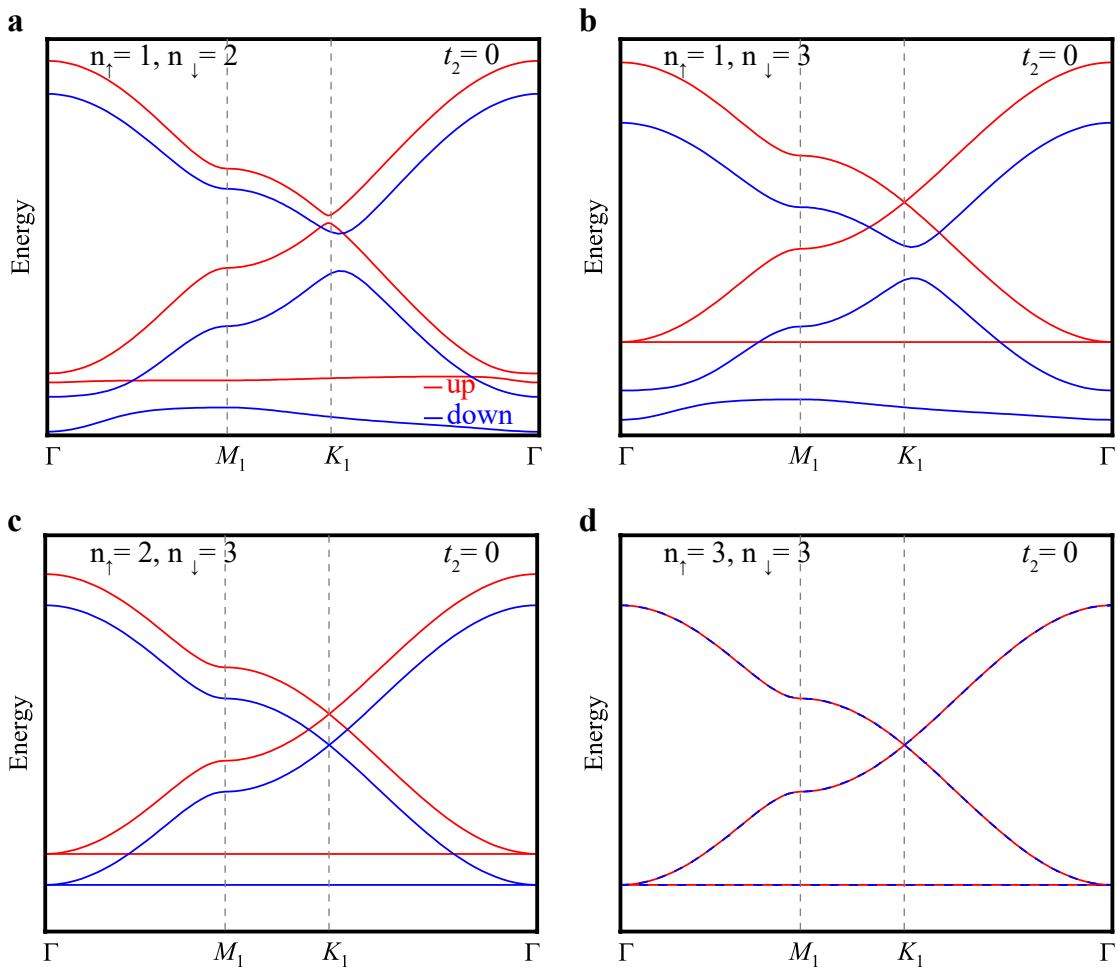
Part 10. The Hubbard mean field calculation of CT lattice band structure

To qualitatively evaluate the effect of electron correction on the CT lattice band structure, we applied the Hubbard mean field method to the CT lattice. The Hubbard mean field Hamiltonian H for the CT lattice is:

$$-t_1 \sum_{\langle i,j\rangle,\sigma} \left(c^{\dagger}_{i,\sigma} c_{j,\sigma}\right) - t_1^{'} \sum_{\langle i,j\rangle,\sigma} \left(c^{\dagger}_{i,\sigma} c_{j,\sigma}\right) - t_2 \sum_{\langle i,j\rangle,\sigma} \left(c^{\dagger}_{i,\sigma} c_{j,\sigma}\right) - t_3 \sum_{\langle i,j\rangle,\sigma} \left(c^{\dagger}_{i,\sigma} c_{j,\sigma}\right) - t_3^{'} \sum_{\langle i,j\rangle,\sigma} \left(c^{\dagger}_{i,\sigma} c_{j,\sigma}\right) - t_4 \sum_{\langle i,j\rangle,\sigma} \left(c^{\dagger}_{i,\sigma} c_{j,\sigma}\right) + U \sum_{i} \left(n_{i,\uparrow}\langle n_{i,\downarrow}\rangle + n_{i,\downarrow}\langle n_{i,\uparrow}\rangle - \langle n_{i,\uparrow}\rangle\langle n_{i,\downarrow}\rangle\right)$$

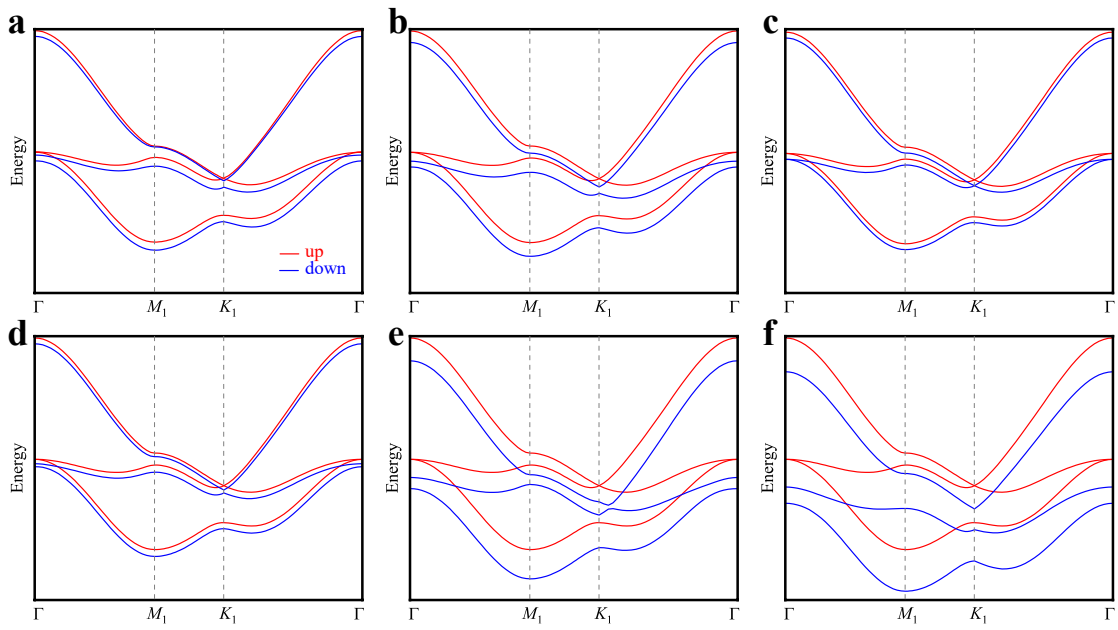
where $c^{\dagger}_{i,\sigma}$ and $c_{j,\sigma}$ are the creation and annihilation operators, respectively, σ denotes the spin, $n_{i,\sigma}$ and $\langle n_{i,\sigma}\rangle$ are the number operator and mean field electron density for spin σ electrons at site i, respectively. Here, U is the Hubbard on-site Coulomb interaction energy, and t_1, $t_1^{'}$, and t_2 (t_3, $t_3^{'}$, and t_4) are the nn (nnn) hopping parameters as defined in the CT lattice. The summation was taken over all spin indices and nn or nnn pairs connected with the specific hopping parameters. The Hubbard mean field band structures are calculated by employing an open-access algorithm developed by Bernard Field.

Supplementary Figure 24 illustrates the typical band structures of the CT lattice Hubbard model with $t_1 = t_1$' and $U = 0.5\ t_1$. The results show that the Hubbard on-site Coulomb interaction can induce a gap opening at the K_1 point. However, it also causes a gap opening at the Γ point, which is not observed in the DFT bands. Given that the Dirac band and the flat band in our DFT calculations are degenerate at the Γ point for both spin channels, we infer that the structural asymmetry of a_1 and a_1' (i. e., $t_1 \neq t_1$') likely plays a more crucial role in the gap opening at the K_1 point.



Supplementary Figure 24. Schematic band structures of the Hubbard mean field model of the CT lattice, in the absence of nnn hopping, with $t_1 = t_1$' and $U = 2\ t_1$. (a-d) Hubbard mean field band structures in the case of $t_2 = 0$ with (a) $n_\uparrow = 1$, $n_\downarrow = 2$; (b) $n_\uparrow = 1$, $n_\downarrow = 3$; (c) $n_\uparrow = 2$, $n_\downarrow = 3$; and (d) $n_\uparrow = 3$, $n_\downarrow = 3$.

We also explored the Hubbard mean field model of the CT lattice, incorporating both nn and nnn hopping and analyzed the band structure's dependence on the on-site Coulomb interaction strength. As shown in Supplementary Figure 25, the discrepancy between DFT and TB bands near the VBM energy range remains unresolved even when the electron correction is included via the mean field Hubbard model.



Supplementary Figure 25. Schematic band structures of the Hubbard mean field model of the CT lattice, in consideration of both nn and nnn hopping, with $t_1 = t_1$'; $t_2 = 0.3\ t_1$; $t_3 = t_4 = 0.85\ t_1$; and t_3' = -0.2 t_1. (a-c) Hubbard mean field band structures in the case of $U = t_1$ with (a) $n_\uparrow = 1$, $n_\downarrow = 2$; (b) $n_\uparrow = 1$, $n_\downarrow = 3$; and (c) $n_\uparrow = 2$, $n_\downarrow = 3$. (d-f) Hubbard mean field band structures in the case of $n_\uparrow = 1$, $n_\downarrow = 3$ with (d) $U = 0.5\ t_1$, (e) $U = 2\ t_1$; and (f) $U = 3\ t_1$.

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